🛡️ Cybersecurity & Ethical Hacking Internship Report

Organization: Ansh Infotech

Role: Cybersecurity Intern

Duration: January 2026 - Present

Focus: Linux Administration, Networking Fundamentals, Vulnerability Assessment, Social Engineering

📌 Executive Summary

This repository documents my daily learning progress, lab setups, and practical exercises during my internship. The training spans from core Linux system administration to offensive security techniques like social engineering and network exploitation.

📅 Phase 1: Linux Fundamentals & System Administration

Dates: Jan 15, 2026 – Jan 28, 2026

🔹 Learning Modules

1. **Linux Architecture:**
    - Explored distributions: **Kali Linux** (Offensive Security), **Ubuntu**, and **Debian**.
    - Understood the Linux File Hierarchy Structure (root /, /home, /etc, /var).
    - Desktop Environments.
2. **Essential CLI Commands:**
    - **File Ops:** touch (create), cat (read), cp (copy), mv (move), rm (remove).
    - **Editors:** Using nano for terminal-based text editing.
    - **System Info:** uname -a (kernel info), whoami, ls -h (human-readable list).
3. **User & Permission Management (Crucial for Security):**
    - **Permissions:** Mastered the **rwx** (Read-4, Write-2, Execute-1) system.
    - **Commands:**
        - chmod: Changing file modes (e.g., chmod +x script.sh to make executable).
        - chown: Changing file ownership.
        - sudo vs su: Superuser execution vs. switching user context.
4. **Package & Process Management:**
    - Managed software using apt (Advanced Package Tool): update, install, purge.

- **Monitoring:** Used top and ps to track real-time system resources.

📸 Practical Evidence

Practicing file permission commands and process monitoring.

Permission	Number
No permission	0
Execute (x)	1
Write (w)	2
Write & Execute (wx)	3
Read (r)	4
Read & Execute (rx)	5
Read & Write (rw)	6
Read, Write, & Execute (rwx)	7

Permissions	Alphabetic	Numeric	Description
---	`---`	0	No permissions.
--x	`--x`	1	Execute only.
-w-	`-w-`	2	Write only.
-wx	`-wx`	3	Write and execute.
r--	`r--`	4	Read only.
r-x	`r-x`	5	Read and execute.
rw-	`rw-`	6	Read and write.
rwx	`rwx`	7	Read, write, and execute.

📅 Phase 2: Vulnerability Assessment & Lab Setup

Date: Jan 29, 2026

🔹 Lab Configuration

Transitioned to an offensive mindset by setting up a safe, isolated environment for testing exploits.

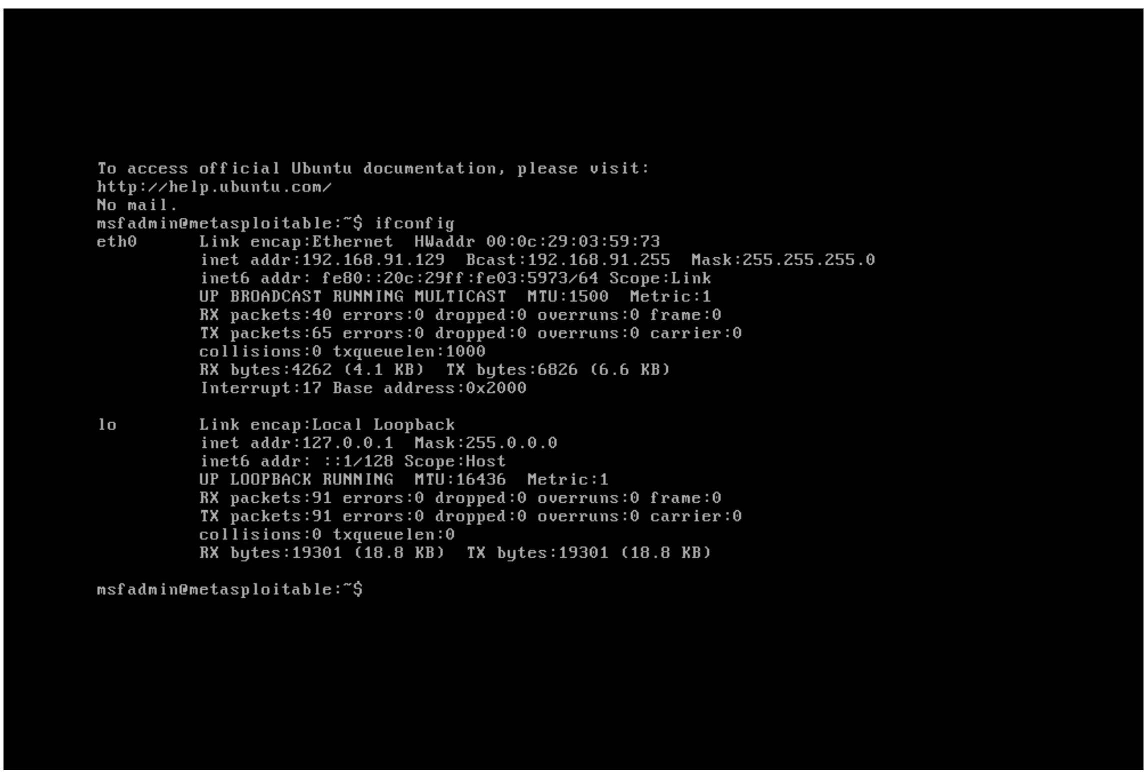
- **Target Machine:** Metasploitable (intentionally vulnerable Linux VM).
- **Attacker Machine:** Kali Linux.

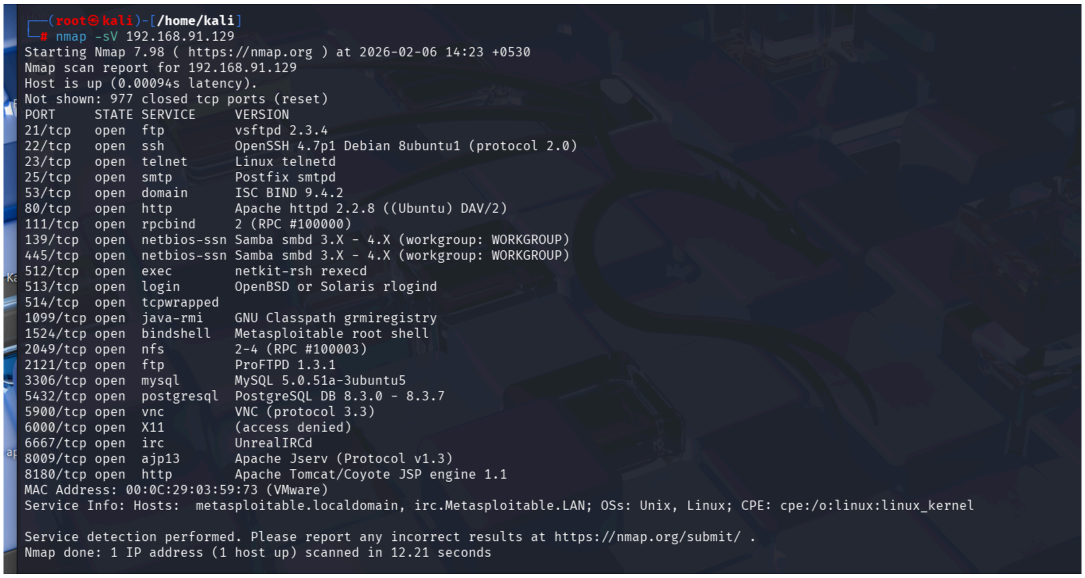
🔹 Tools & Techniques

- **NMAP (Network Mapper):**
  - Used for network discovery and security auditing.
  - Scanned targets to identify open ports and services.
- **Exploitation Basics:**
  - Connected to the vulnerable lab using **FTP** and **msfconsole** (Metasploit Framework).
- **Phishing Simulation:**
  - Deployed **Zphisher** to understand credential harvesting attacks.

📸 Practical Evidence

Setup of Metasploitable lab and initial port scanning.





```
┌──(root㉿kali)-[/home/kali]
└─# nmap -sV 192.168.91.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-06 14:23 +0530
Nmap scan report for 192.168.91.129
Host is up (0.00094s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:03:59:73 (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.21 seconds
```

📅 Phase 3: Networking Core Concepts

Dates: Jan 30, 2026 – Feb 4, 2026

🔹 Theory & Architecture

A solid understanding of networking is the backbone of cybersecurity.

1. **Network Topologies:**
   - **PAN:** Personal Area (Bluetooth/NFC).
   - **LAN/WLAN:** Local Area (Home/Office Wi-Fi).
   - **MAN/WAN:** Metropolitan & Wide Area (Inter-city/Internet).
2. **The OSI Model (7 Layers):**
   - Physical, Data Link, Network, Transport, Session, Presentation, Application.
   - *Key Takeaway:* Understanding where data exists (cable vs. IP packet vs. HTTP request) helps in diagnosing attacks.
3. **Addressing & Protocols:**
   - **MAC Address:** 48-bit physical hardware address (Hexadecimal).
   - **IP Address:** Logical address (IPv4 vs. IPv6).
   - **Transmission Modes:** Unicast (1-to-1), Multicast (1-to-many), Broadcast (1-to-all).
4. **Ports & Services Table:**
   - **FTP:** 20/21 (File Transfer)
   - **SSH:** 22 (Secure Shell)
   - **HTTP/HTTPS:** 80/443 (Web)
   - **DNS:** 53 (Domain Name System)

📅 Phase 4: Ethical Hacking - Social Engineering

Date: Feb 6, 2026

🔹 Human-Centric Vulnerabilities

Started the "Ethical Hacking" module, focusing on the weakest link in security: the human element.

🔹 Attack Vectors Studied

1. **Phishing:** Fraudulent communications to induce users to reveal sensitive info.
2. **Pretexting:** Fabricating scenarios (e.g., posing as HR or IT) to steal data.
3. **Baiting:** Using physical media (USB drives) or free downloads to entice victims.
4. **Tailgating/Piggybacking:** Following authorized personnel into restricted areas.
5. **Scareware:** Malicious software confusing users into visiting malware sites.

🔹 Tools Used

- **Zphisher:** For automated phishing templates.
- **ALHacking:** Comprehensive hacking tool suite.

📸 Practical Evidence

Running Zphisher for educational phishing simulation.

📅 Phase 5: Real-World Reconnaissance & SSH Exploitation Testing

Dates: Feb 11, 2026 – Feb 12, 2026

🔹 Target Analysis & Information Gathering (Feb 11)

- **Target:** `gndec.ac.in` `(175.176.187.102)`
- **Reconnaissance:** Conducted active scanning using Nmap (`nmap -sV -sC`) to map out open ports, services, and OS fingerprints.
- **Findings:** Identified legacy software running on the live target, specifically OpenSSH 7.2p2 and Apache httpd 2.4.18, indicating an older Ubuntu 16.04 server environment.

🔹 Vulnerability Assessment & SSH Exploitation (Feb 12)

SSH Username Enumeration (CVE-2018-15473): * Attempted automated enumeration using Metasploit (`auxiliary/scanner/ssh/ssh_enumusers`).

- Identified that high network latency (2.2s) caused a "false positive storm" due to the tool's integer-based threshold limitations.

Manual Timing Analysis: * Bypassed automated tool limitations by manually testing SSH response times (`time ssh`).

- Discovered an "inverted" timing pattern where valid users (e.g., `root`) responded significantly faster (0.52s) than nonexistent users (3.73s). Successfully confirmed the vulnerability with a clear ~3.2s timing delta.

SSH Brute-Force Testing: * Utilized Metasploit's `auxiliary/scanner/ssh/ssh_login` module to launch a dictionary attack against identified users.

- Monitored the server's response to high-frequency login attempts to evaluate the effectiveness of Intrusion Prevention Systems (e.g., Fail2Ban), noting the server permitted over 120 rapid sequential attempts without immediate IP blocking.

📅 Phase 6: Mobile Application Security & Static Analysis

Date: Feb 13, 2026

🔹 Mobile Security Assessment

Conducted a comprehensive Static Application Security Testing (SAST) audit on a third-party modified Android application to identify privacy violations, hardcoded secrets, and vulnerability patterns.

- **Target Application:** `InstaPro_v14.25.apk` (Modded Instagram Client)
- **Package Name:** `com.instapro.android`
- **Tool Used: MobSF (Mobile Security Framework) v4.4.5**

🔹 Key Findings (Static Analysis)

Security Score:

- The application received a **Security Score of 50/100 (Grade B)**, indicating Medium Risk.

Dangerous Permissions:

- Analysis revealed excessive permission requests that pose privacy risks, including:
    - `android.permission.READ_CONTACTS` (Read user contacts)
    - `android.permission.RECORD_AUDIO` & `CAMERA` (Surveillance risk)

  - `android.permission.READ_CALL_LOG` & `READ_SMS` (Sensitive communication data)
  - `android.permission.ACCESS_FINE_LOCATION` (Precise GPS tracking)

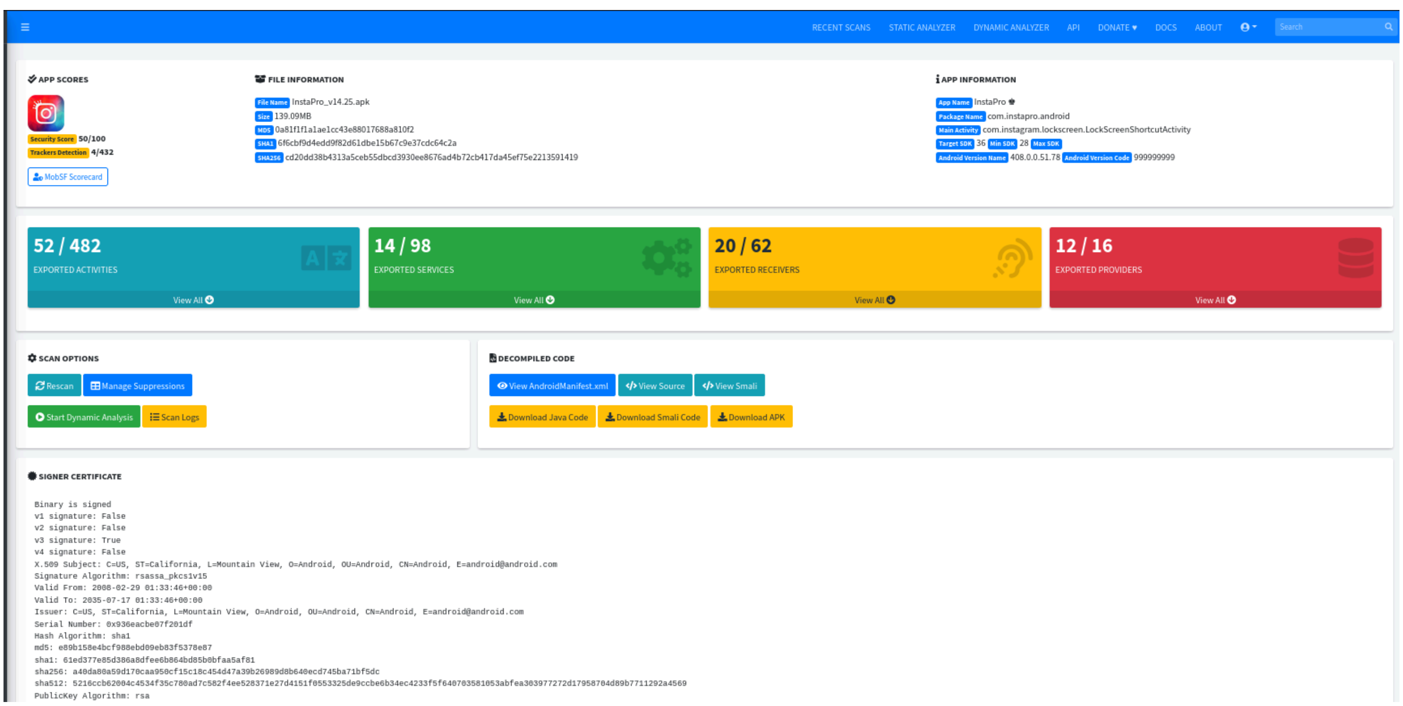
Hardcoded Secrets & Information Leakage:

- **API Keys:** Found exposed `google_api_key` and Firebase configuration URLs (`firebase_database_url`) in the source code.
- **Log Leakage:** Detected source code explicitly logging sensitive information, violating OWASP MSTG-STORAGE-3.

Code Vulnerabilities:

- **Cryptographic Weakness:** Identified the use of `CBC` encryption mode with `PKCS5/PKCS7` padding, which is vulnerable to padding oracle attacks (High Severity).
- **SQL Injection:** Found raw SQL queries executing untrusted user input (CWE-89).
- **Insecure Randomness:** Use of insufficient random number generators (CWE-330).

📸 Practical Evidence



Generated a full PDF report via MobSF highlighting the security score, tracker detection (4/432), and the manifest analysis of the `InstaPro` APK.

📅 Phase 7: Hands-on Exploitation (Hack The Box)

Date: Feb 16, 2026

🔹 CTF / Lab Challenges

Started practical machine exploitation on the Hack The Box (HTB) platform to reinforce scanning and enumeration skills.

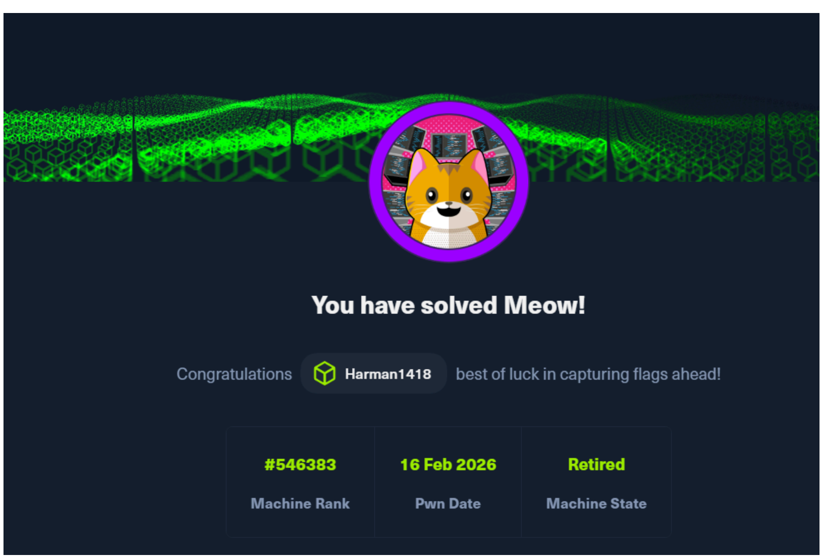
Target 1: Meow (Linux)

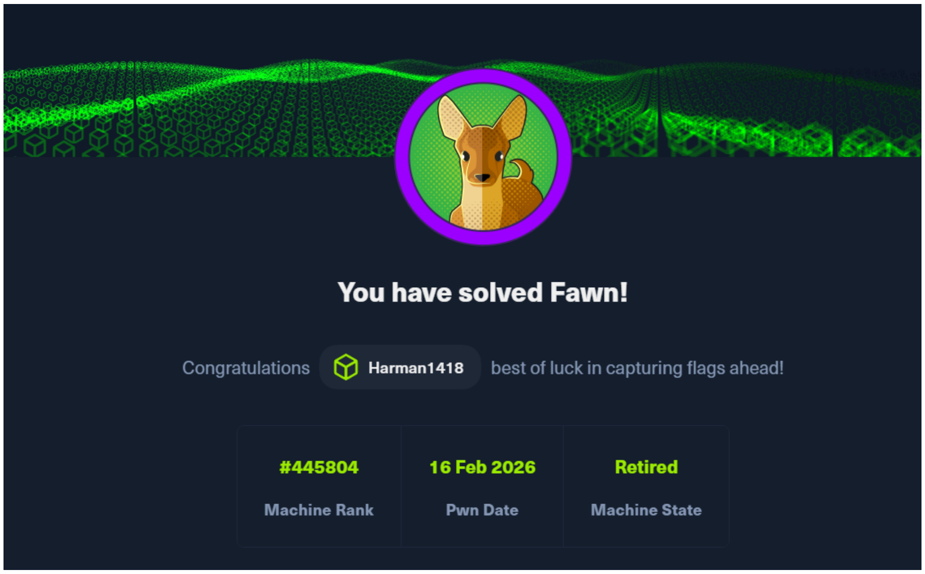
- **Difficulty:** Very Easy.
- **Objective:** Gain root access via Telnet.
- **Process:**
    - Connected via OpenVPN.
    - Scanned target using `nmap`. Identified Port 23 (Telnet) as open.
    - Attempted connection using `telnet <IP>`.
    - Exploited weak configuration by logging in as `root` with no password.
    - Retrieved the `root.txt` flag.

Target 2: Fawn (Linux)

- **Difficulty:** Very Easy.
- **Objective:** Data exfiltration via FTP.
- **Process:**
    - Scanned target using `nmap`. Identified Port 21 (FTP) as open.
    - Attempted `ftp <IP>` and successfully logged in using the `anonymous` user.
    - Used `ls` to list files and `get flag.txt` to download the flag.

📸 Practical Evidence





📅 Phase 8: Metasploitable Pentesting & Windows Setup

Date: Feb 17, 2026

🔹 Lab Penetration Testing (Metasploitable)

Continued penetration testing on the Metasploitable lab environment, focusing on remote access protocols.

Target: Metasploitable Linux VM. **Tools:** `msfconsole` (Metasploit Framework). **Protocols:** SSH (Port 22), Telnet (Port 23).

Exploitation Process:

- **Objective:** Perform dictionary attacks to gain credentials for remote access services.
- **Issue Encountered:** During the attack execution with `msfconsole`, the scanner failed to read the `user.txt` file correctly. The output showed: `[1] authentication failed ":<pass>"` instead of the expected format: `[1] authentication failed "<user>:<pass>"`
- **Troubleshooting & Fix:**
  - Analyzed the module configuration and input format.
  - **Solution:** Created a combined dictionary file named `userpass.txt` where usernames and passwords were separated by a space (format: `username password`).
  - **Result:** Re-running the attack with `userpass.txt` resolved the parsing error, allowing successful credential testing on both SSH and Telnet services.

🔹 Windows Environment Setup

- **Objective:** Prepare a secure testing ground for advanced post-exploitation techniques.
- **Action:** Successfully installed **Windows 11 Enterprise** on VMware.
- **Future Plan:** This environment will be used to test credential dumping tools like **Mimikatz** in an upcoming session.

📅 Phase 9: Post-Exploitation & Network Revision

Date: Feb 18, 2026

🔹 Advanced Credential Dumping (Mimikatz)

Focused on privilege escalation and credential extraction on Windows environments.

Windows 11 Enterprise (Attempt):

- **Objective:** Extract credentials from memory using Mimikatz.
- **Challenges:** Encountered enhanced security protections (Credential Guard/LSA Protection) preventing access.
- **Troubleshooting Steps:**
  - Attempted to run via `nsexec` to gain SYSTEM privileges.
  - Modified Registry permissions via `regedit` to disable LSA protections.
- **Outcome:** Despite troubleshooting, modern Windows 11 defenses prevented successful extraction in the lab environment.

Windows 10 Enterprise (Success):

- **Action:** Migrated testing to a Windows 10 Enterprise VM.
- **Outcome:** Successfully executed Mimikatz commands (`privilege::debug`, `sekurlsa::logonpasswords`) to retrieve NTLM hashes and cleartext credentials.

🔹 Network Security Revision

Revisited core networking concepts to solidify the theoretical foundation for advanced attacks.

- **Topics Reviewed:** OSI Model layers, common ports (21, 22, 80, 443, 445), and TCP/IP handshake.
- **Practical Exercise:** Conducted a Denial of Service (DoS) simulation using **hping3** to understand packet flooding mechanics (`hping3 -S --flood -V -p 80 <Target IP>`).

🔹 Homework: TryHackMe Lab

- **Module:** Post-Exploitation Basics.
- **Key Learnings:** Enumeration after compromise, maintaining access, and dumping system hashes using tools like **Hashcat** and **John the Ripper**.

📅 *Phase 10: Web Application Logic Testing*

Date: *Feb 19, 2026*

🔹 *Live Target Vulnerability Assessment*

Performed a business logic security test on a live e-commerce environment to identify parameter tampering vulnerabilities.

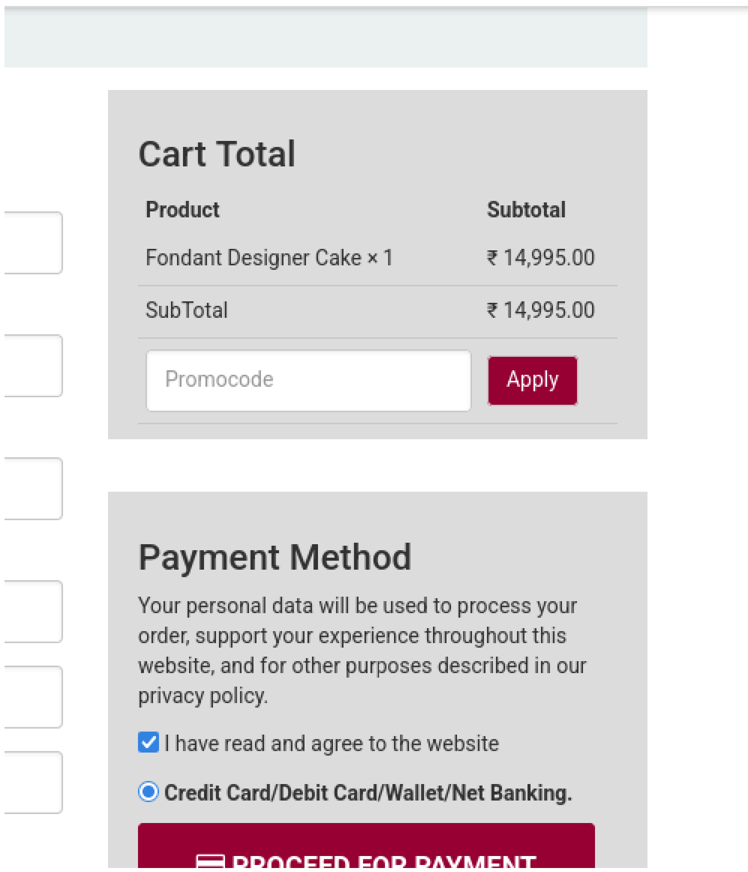
- ***Target:*** `www.ogdmart.com`
- ***Vulnerability:*** *Parameter Tampering (Price Manipulation).*
- ***Tools Used:*** *Burp Suite (Proxy Interception).*

🔹 *Methodology*

1. ***Interception:*** *Configured browser proxy to route traffic through Burp Suite.*
2. ***Transaction Analysis:*** *Initiated a checkout process for a high-value item ("Fondant Designer Cake").*
3. ***Parameter Identification:*** *Analyzed the HTTP POST request body during the payment initiation phase.*
   - ***Original Value:*** `cart_total=14995`
4. ***Tampering:*** *Modified the* `cart_total` *parameter from* `14995` *to* `5` *in the intercepted request before forwarding it to the server.*
5. ***Verification:***
   - ***Result:*** *The application accepted the modified value. The payment gateway page reflected the tampered price of **₹5** instead of the original **₹14,995**.*
   - ***Ethical Note:*** *The process was halted at the payment gateway page. No actual transaction was completed to strictly adhere to ethical hacking guidelines.*

📸 *Practical Evidence*

Screenshots demonstrating the request interception and the reflected price change on the payment page.



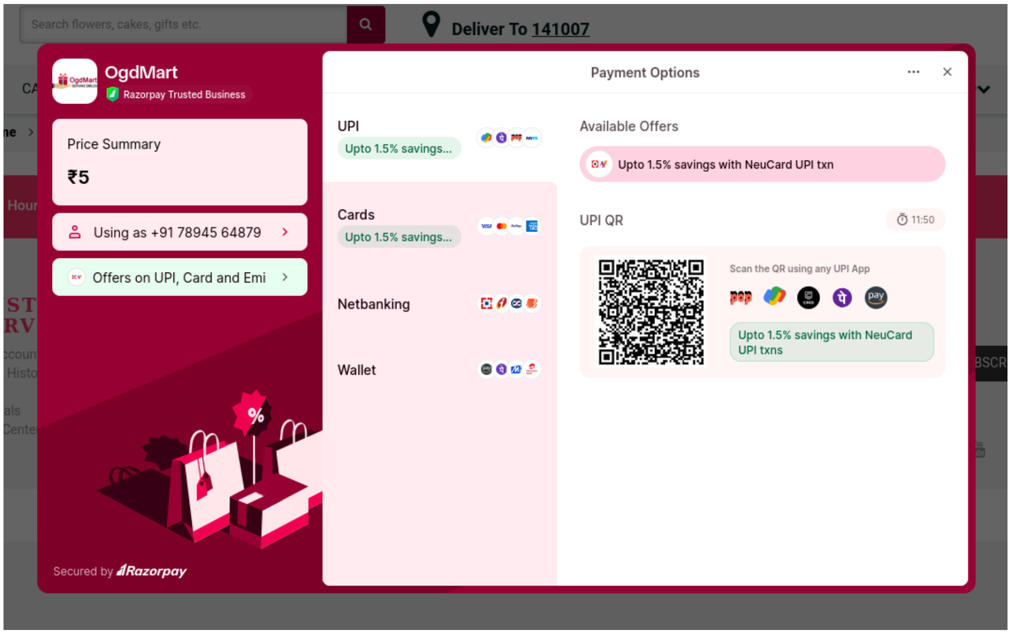
Before Tampering:

```
Te: trailers

txtBillingAddressFname=Burp&txtBillingAddressLname=Suite&txtBillingAddressEmailAddress=haaefa%40gmail.com&txtBillingAddressPhone=9876543210&txtBillingAddressCompany=&txtBillingAddressLine1=
11231%2Cdsfd%2Cdsfs&txtBillingAddressLine2=&txtBillingAddressLine3=&txtBillingTown=delhi&txtBillingState=delhi&txtBillingPostcode=141007&txtBillingCountry=INDIA&txtOrderNotes=&password=&
txtShippingAddressFname=Burp&txtShippingAddressLname=Suite&txtShippingAddressEmailAddress=haaefa%40gmail.com&txtShippingAddressPhone=9876543210&txtShippingAddressLine1=11231%2Cdsfd%2Cdsfs&
txtShippingAddressLine2=&txtShippingAddressLine3=&txtShippingTown=delhi&txtShippingState=delhi&txtShippingPostcode=141007&txtShippingCountry=INDIA&coupon_code=&cart_subt=14995&cart_disc=0&
cart_total=14995&paymenttype=2
```

After Tampering:

```
txtBillingAddressFname=Burp&txtBillingAddressLname=Suite&txtBillingAddressEmailAddress=haaefa%40gmail.com&txtBillingAddressPhone=9876543210&txtBillingAddressCompany=&txtBillingAddressLine1=
11231%2Cdsfd%2Cdsfs&txtBillingAddressLine2=&txtBillingAddressLine3=&txtBillingTown=delhi&txtBillingState=delhi&txtBillingPostcode=141007&txtBillingCountry=INDIA&txtOrderNotes=&password=&
txtShippingAddressFname=Burp&txtShippingAddressLname=Suite&txtShippingAddressEmailAddress=haaefa%40gmail.com&txtShippingAddressPhone=9876543210&txtShippingAddressLine1=11231%2Cdsfd%2Cdsfs&
txtShippingAddressLine2=&txtShippingAddressLine3=&txtShippingTown=delhi&txtShippingState=delhi&txtShippingPostcode=141007&txtShippingCountry=INDIA&coupon_code=&cart_subt=5&cart_disc=0&
cart_total=5&paymenttype=2
```



📅 Phase 11: Web Application Security & Environment Setup

Date: *Feb 20, 2026*

🔹 *Live Target Assessment: OTP Verification*

Conducted an assessment on a live financial portal to understand the implementation and potential weaknesses of One-Time Password (OTP) authentication mechanisms.

- ***Target:*** *`www.instafinancials.com`*
- ***Vulnerability Category:*** *Broken Authentication / Business Logic Flaw (OTP Bypass)*
- ***Tools Used:*** *Burp Suite, FoxyProxy*

🔹 *Methodology & Exploitation Concept*

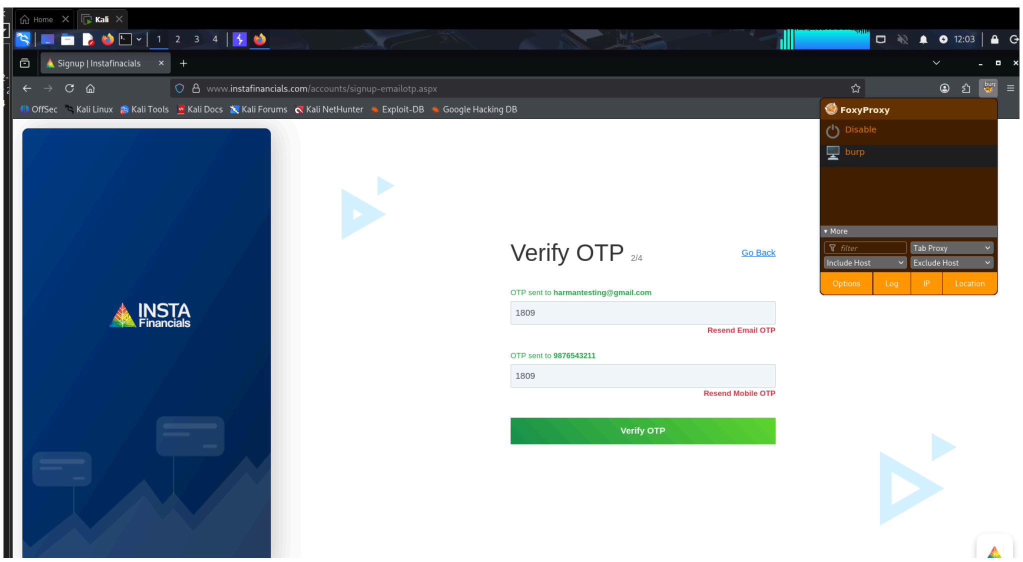
1. ***Traffic Interception:*** *Configured FoxyProxy to route browser traffic through Burp Suite.*
2. ***Triggering Verification:*** *Initiated the account registration process and proceeded to the OTP verification step using testing data.*
3. ***Invalid Submission:*** *Entered an arbitrary, incorrect OTP (e.g., `1809`) to trigger a validation request.*
4. ***Response Manipulation:*** *Intercepted the server's response to the submitted OTP request. Modified the server response data in transit to simulate a successful*

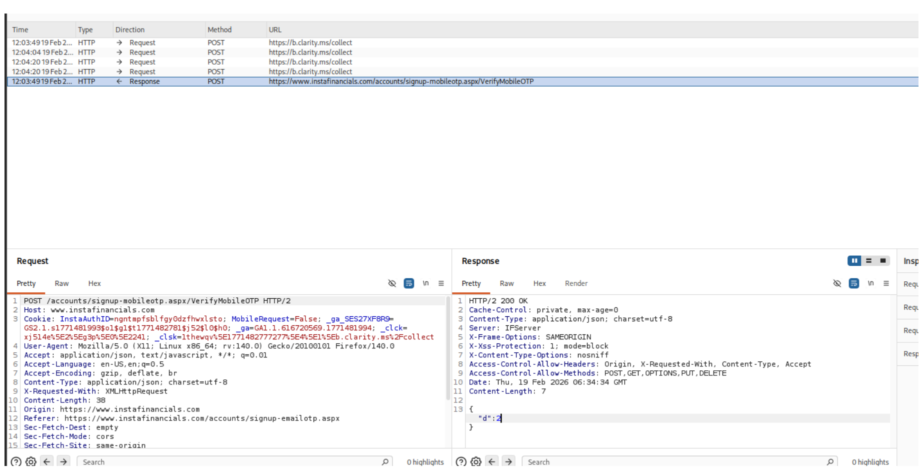
validation status.

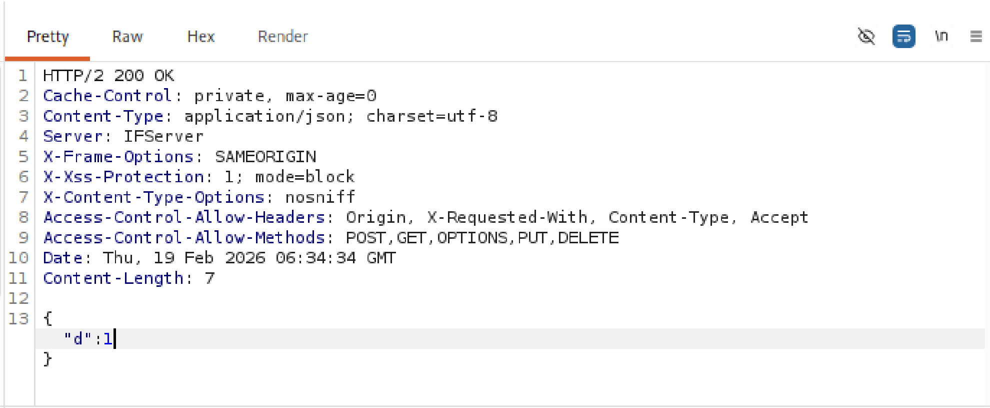
5. ***Verification:*** *Forwarded the manipulated response back to the browser. The frontend application improperly trusted the modified response, bypassing the OTP check and granting access.*
    - ***Ethical Note:*** *Testing was conducted purely for educational purposes to demonstrate the risks of relying on client-side validation rather than strict server-side enforcement. No unauthorized access to sensitive user data was performed.*

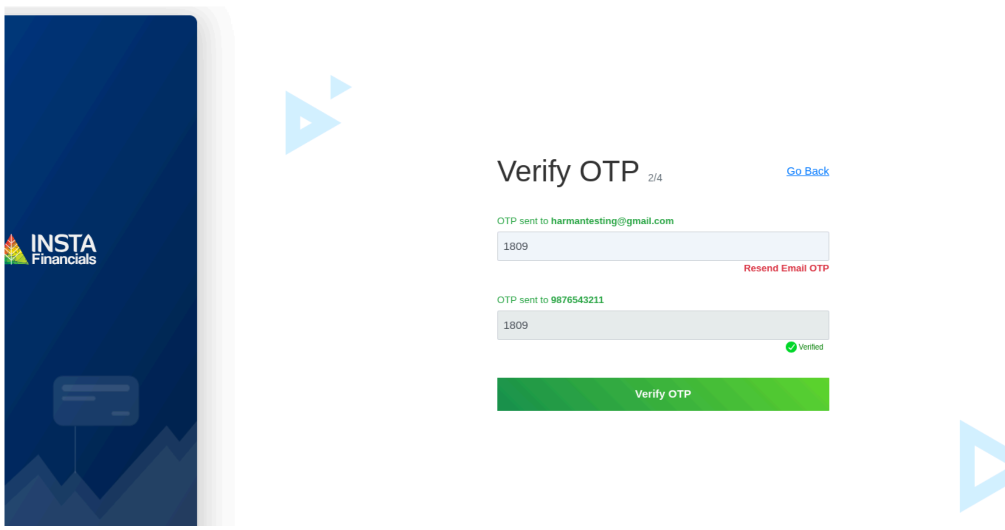
📸 *Practical Evidence*

Screenshots demonstrating the interception of the OTP request and the successful bypass via Burp Suite.









Environment Configuration: WSL & Kali Linux GUI

Expanded the local testing infrastructure by integrating Kali Linux directly into the Windows environment for streamlined access to penetration testing tools.

- ***Technology Used:*** *Windows Subsystem for Linux (WSL 2), Win-KeX.*
- ***Process:***
  - ***WSL Deployment:*** *Enabled the WSL feature on Windows and successfully installed the* ***Kali Linux*** *distribution natively.*
  - ***GUI Integration (Win-KeX):*** *To utilize graphical security tools seamlessly without a standalone VM, installed and configured* ***Win-KeX*** *(Kali Desktop Experience for Windows).*
  - ***Execution:*** *Initialized the graphical desktop environment using the* `kex` *command, establishing a persistent, low-overhead Kali GUI running parallel to the Windows host.*
- ***Outcome:*** *Achieved a fully functional, integrated Kali Linux desktop environment, significantly improving the speed and efficiency of the overall testing workflow.*

📅 *Phase 12: Active Reconnaissance & Portfolio Development*

Dates: *Feb 22, 2026 – Feb 23, 2026*

🔹 *Professional Portfolio Development (Feb 22)*

- ***Objective:*** *Establish a professional online presence to showcase cybersecurity projects, vulnerability assessments, and technical write-ups.*
- ***Action:*** *Developed and deployed a personal portfolio website hosted at* `https://harmanjot-singh.vercel.app/`

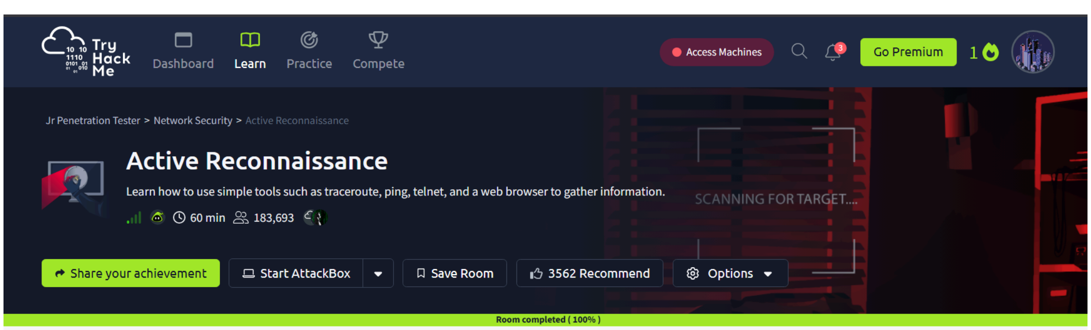
🔹 *Active Reconnaissance Lab (Feb 23)*

Completed the ***Active Reconnaissance*** *room on the TryHackMe platform to reinforce network enumeration and information gathering techniques.*

- ***Tools & Commands Practiced:***
  - **`ping`** ***/*** **`ping -c`** ***/*** **`ping -n`*****:*** *Utilized ICMP echo requests to verify host availability on both Linux/macOS and Windows systems.*
  - **`traceroute`** ***/*** **`tracert`*****:*** *Mapped the network path and routing hops to target systems to understand network topology.*
  - **`telnet`*****:*** *Conducted manual port checking and banner grabbing on specific ports.*
  - **`nc`** ***(Netcat):*** *Practiced establishing connections as a client (*`nc <IP> <PORT>`*) and setting up listeners as a server (*`nc -lvnp <PORT>`*).*
- ***Outcome:*** *Achieved 100% completion of the room, successfully demonstrating the ability to gather actionable intelligence using simple, built-in system tools.*

📸 *Practical Evidence*

TryHackMe room completion-



📅 *Phase 13: Incident Response & Advanced Web Security*

Date: *Feb 24, 2026*

🔹 *Incident Response (TryHackMe)*

Initiated the study of defensive cybersecurity frameworks, focusing on the first phase of the Incident Response lifecycle.

- ***Module:*** *Incident Response - Phase 1: Preparation (TryHackMe).*
- ***Key Learnings:*** ** Understanding the core components of an Incident Response Plan (IRP).*
    - *The importance of establishing policies, training response teams, and preparing technical tools before a security breach occurs.*
    - *Evaluating business impact and configuring logging/monitoring solutions to ensure readiness.*

🔹 *Live Target Assessment: OTP Verification Flaw*

Conducted a secondary real-world assessment to reinforce concepts around broken authentication and client-side trust issues.

- ***Target:*** `www.bakingo.com`
- ***Vulnerability Category:*** *Broken Authentication / Business Logic Flaw (OTP Bypass)*
- ***Tools Used:*** *Burp Suite*
- ***Methodology:*** *1. Triggered the OTP verification process during the login/registration flow. 2. Intercepted the validation request using Burp Suite Proxy. 3. Manipulated the server's response to the client, effectively forcing a "Success" condition on an invalid OTP entry. 4. The application improperly trusted the manipulated response and bypassed the authentication barrier.*
- ***Ethical Note:*** *This assessment was performed strictly for educational practice and concept reinforcement. The activity was halted post-verification, ensuring no unauthorized account takeovers or data alterations occurred.*

🔹 *Continuous Learning & Upskilling*

To stay ahead of modern threat landscapes, began supplementing foundational knowledge with advanced paradigms.

- ***Course:*** *"AI for CyberSecurity and Bug Bounty Hunting" (Udemy).*
- ***Objective:*** *Learn to leverage Artificial Intelligence, Large Language Models (LLMs),*

and Machine Learning techniques to automate reconnaissance, identify complex vulnerability patterns, and optimize bug bounty workflows.

🛠️ *Technology Stack*

- ***OS:*** *Kali Linux, Ubuntu, Windows 10/11 Enterprise, WSL 2*
- ***Networking:*** *OSI Model, TCP/IP, IPv4/IPv6*
- ***Tools:*** *Nmap, Metasploit, Zphisher, MobSF, Nano, Apt, OpenVPN, Mimikatz, hping3, Burp Suite, FoxyProxy, Win-KeX*
- ***Virtualization:*** *VMware/VirtualBox (for Labs)*

📝 *Future Scope*

- *Advanced Packet Analysis with* ***Wireshark****.*
- *Python scripting for automating penetration tests.*
- *Deep dive into Web Application Security (OWASP Top 10).*

Internship Log maintained by Harmanjot Singh